FSD REVIEW 1

NeuroAccess

Cognitive & Vision-Accessible Content Adapter

THE PROBLEM

The Digital Accessibility Gap

Modern web design relies on dense text, intrusive ads, and complex layouts.

This creates severe navigation barriers for neurodivergent and visually impaired users.

20%

Estimated global population with neurodivergent conditions.

PROPOSED SOLUTION

Autonomous Content Adaptation

AI Simplification

Restructures complex paragraphs instantly.

Audio Assistant

Synchronized text-to-speech engine.

Visual Interface

Customizable high-contrast themes.

GLOBAL IMPACT

SDG Alignment



SDG 3: Good Health & Well-being

Promoting mental well-being and significantly reducing cognitive fatigue for neurodivergent individuals by creating stress-free, highly accessible digital reading environments.

WORKFLOW

System Architecture



CORE LOGIC

Backend Infrastructure

A robust Node.js and Express server orchestrating data extraction and AI processing.

Puppeteer Extraction

Headless browser dynamically traverses the DOM to scrape article text while filtering out ads and navigational bloat.

Gemini AI Integration

API endpoints stream scraped text to Google Gemini for structural simplification and neuro-inclusive rephrasing.

RESTful Routing

Express handles cross-origin requests, validates URLs, and efficiently pipes processed JSON payloads back to the client.

DATA PERSISTENCE

Database Engineering

MongoDB schemas engineered for low-latency profile retrieval.

Utilizes the Mongoose ODM to ensure strict schema validation and rapid state synchronization for personalized accessibility preferences across sessions.

Collection	Key Fields	Type
Profiles	`theme`	String
Profiles	`fontSize`	Number
Profiles	`speechRate`	Number
Profiles	`fontFamily`	String

USER INTERFACE

Front-End Architecture

- **React 18 & Vite:** Component-driven architecture ensuring fast re-renders and modular, highly maintainable UI layout.
- **Context API State:** Real-time CSS variable injection for instant high-contrast theme and dynamic typography switching.
- **Web Speech API Sync:** Custom sequential TTS engine featuring skimmable audio tracking and precise word-level DOM highlighting.
- **Async Integration:** Robust fetch logic with graceful error handling and loaders bridging the React UI with the Node backend.

Live Demonstration

Implementation Walkthrough & Q&A